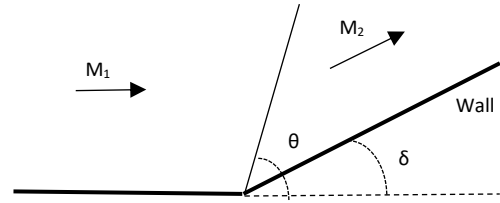
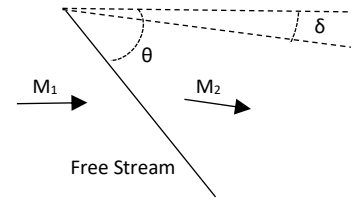


Oblique Shock Wave



θ = shock wave angle

δ = ramp angle (physical or virtual)



Given P_2/P_1

$$\frac{P_2}{P_1} = \frac{2\gamma}{\gamma+1} M_1^2 \sin^2 \theta - \frac{\gamma-1}{\gamma+1} \Rightarrow \sin \theta = \sqrt{\frac{\left(\frac{P_2}{P_1} + \frac{\gamma-1}{\gamma+1}\right)}{\frac{2\gamma}{\gamma+1} M_1^2}}$$

$$\cot \delta = \tan \theta \frac{(\gamma+1)M_1^2 - 2(M_1^2 \sin^2 \theta - 1)}{2(M_1^2 \sin^2 \theta - 1)}$$

Given δ

$$\frac{\cot \delta}{\tan \theta} = \frac{(\gamma+1)M_1^2 - 2(M_1^2 \sin^2 \theta - 1)}{2(M_1^2 \sin^2 \theta - 1)}$$

Evaluate θ by Newton-Raphson method

$$F = -2 \cot \delta (M_1^2 \sin^2 \theta - 1) + (\gamma+1)M_1^2 \tan \theta - 2M_1^2 \tan \theta \sin^2 \theta - 2 \tan \theta = 0$$

$$\frac{dF}{d\theta} = -\frac{2}{\tan \delta} M_1^2 2 \sin \theta \cos \theta + (\gamma+1)M_1^2 \sec^2 \theta - 2M_1^2 (\sec^2 \theta - \cos^2 \theta + \sin^2 \theta) + 2 \sec^2 \theta$$

$$\theta^{i+1} = \theta^i - \frac{F^i}{(dF/d\theta)^i}$$

Then evaluate other properties,

$$M_2^2 = \frac{1}{\sin^2(\theta - \delta)} \frac{1}{\frac{\tan \theta}{\tan(\theta - \delta)} \frac{(\gamma+1)}{2} - \frac{(\gamma-1)}{2}}$$

$$\frac{T_2}{T_1} = \frac{[2\gamma M_1^2 \sin^2 \theta - (\gamma-1)][(\gamma-1)M_1^2 \sin^2 \theta + 2]}{(\gamma+1)^2 M_1^2 \sin^2 \theta}$$

$$\frac{\rho_2}{\rho_1} = \frac{(\gamma+1)M_1^2 \sin^2 \theta}{(\gamma-1)M_1^2 \sin^2 \theta + 2}$$

Derivative of F function

$$\frac{d(\operatorname{tg} \theta)}{d \theta} = \frac{d\left(\frac{\operatorname{sen} \theta}{\cos \theta}\right)}{d \theta} = \sec^2 \theta$$

$$\frac{d \operatorname{sen}^2 \theta}{d \theta} = 2 \operatorname{sen} \theta \cos \theta$$

$$\operatorname{tg} \theta \operatorname{sen}^2 \theta = \frac{\operatorname{sen} \theta}{\cos \theta} \operatorname{sen}^2 \theta = (1 - \cos^2 \theta) \frac{\operatorname{sen} \theta}{\cos \theta} = \operatorname{tg} \theta - \operatorname{sen} \theta \cos \theta$$

$$\frac{d \operatorname{sen} \theta \cos \theta}{d \theta} = \cos^2 \theta - \operatorname{sen}^2 \theta = 1 - 2 \operatorname{sen}^2 \theta$$

$$\begin{aligned} \frac{dF}{d \theta} = & -\frac{2}{\operatorname{tg} \delta} M_1^2 2 \operatorname{sen} \theta \cos \theta + (\gamma + 1) M_1^2 \sec^2 \theta \\ & - 2 M_1^2 (\sec^2 \theta - \cos^2 \theta + \operatorname{sen}^2 \theta) + 2 \sec^2 \theta \end{aligned}$$

Normal Shock Wave

$$M_2^2 = \frac{M_1^2 + \frac{2}{(\gamma - 1)}}{\frac{2\gamma}{(\gamma - 1)} M_1^2 - 1}$$

$$\frac{P_2}{P_1} = \frac{2\gamma}{(\gamma + 1)} M_1^2 - \frac{(\gamma - 1)}{(\gamma + 1)}$$

$$\frac{T_2}{T_1} = \frac{P_2}{P_1} \left[\frac{(\gamma - 1)}{(\gamma + 1)} + \frac{2\gamma}{(\gamma + 1)} M_1^2 \right]$$

$$\frac{\rho_2}{\rho_1} = \frac{(\gamma + 1) M_1^2}{2 + (\gamma - 1) M_1^2}$$